# Conditional Generation with Classically Trainable IQP Circuits

Oriol Balló-Gimbernat[1,2,3] and Emiliano Tolotti[4]

[1]Eurecat, Centre Tecnològic de Catalunya, Barcelona, Spain
[2]Centre de Visió per Computador (CVC), Barcelona, Spain
[3]Universitat Autònoma de Barcelona (UAB), Barcelona, Spain
[4]Department of Information Engineering and Computer Science, University of Trento, Via Sommarive 9, 38123 Trento, Italy

**Conditional quantum generative models usually make the sampling circuit depend on the requested condition, which complicates the "train classically, deploy quantumly" program developed for instantaneous quantum polynomial-time (IQP) Born machines. We instead encode the condition only in a product-state input. One learned phase vector and one compiled commuting circuit then define the whole conditional family, while a character-kernel maximum mean discrepancy objective and its gradients remain accessible through classical Pauli- observable estimators. This gives a particularly simple conditional extension of classically trainable IQP generation without claiming classical access to the full output distribution. Gaussian Hamming-weight targets first isolate continuous peak control and interpolation. A Curie–Weiss Gibbs family gives a more demanding physical example: changing inverse temperature transforms a central magnetization profile into an ordered bimodal one. For this second task we use an unrestricted fixed 17-qubit circuit, measure an eight-qubit visible marginal, and characterize how fidelity changes as zero to nine ancillas are marginalized.**

**Across 8–19 qubits and several target widths, the conditioned circuit moves the generated Hamming-weight profile with the requested peak and consistently outperforms an otherwise identical model that receives no condition. This behavior extends to peak locations not seen during training, demonstrating within-range interpolation. Separate circuits trained for individual peak locations are usually more accurate, quantifying the fidelity cost of using one shared circuit. These results establish product-state conditioning as a practical route from classically trainable IQP generation to conditional generation on this benchmark.**

**Curie–Weiss numerical claims remain disabled until all 180 declared fits and exact visible-marginal evaluations are complete.**

Oriol Balló-Gimbernat: oriol.ballo@eurecat.org
Emiliano Tolotti: emiliano.tolotti@unitn.it

## 1 Introduction

Conditional generative models represent a family $q_\theta(x \mid c)$ rather than a single probability law. Quantum circuit Born machines (QCBMs) encode discrete distributions directly in the measurement statistics of parameterized quantum states and have been studied with likelihood-free, kernel, and divergence-based objectives [1–4]. Conditional QCBMs have also encoded external variables by data-reuploading rotations or additional condition qubits [5, 6]. Those works establish that quantum generators can be conditioned; the unresolved question here is whether conditioning can be added to the recently developed *classically trained* IQP Born machine without changing its compiled sampling circuit or its trained phases from one condition to another.

The distinction between training and sampling is central to this question. For commuting circuits, selected observables can be estimated classically under structural assumptions even though approximate sampling from other IQP families is linked to complexity-theoretic hardness [7–12]. Recent IQP-QCBM work turns that separation into a concrete "train classically, deploy quantumly" program: an MMD objective written in Pauli-$Z$ characters is optimized on classical hard-

ware before the learned circuit is sampled on a quantum device [13, 14]. Our use of *classically trainable* refers only to this observable-estimation interface; it does not assert efficient classical sampling from the learned distribution.

Our construction encodes a scalar condition by broadcast $R_Y$ rotations, then applies one complete two-local IQP circuit. The product input preserves the tractable reference distribution required by the Pauli estimator, while the Hamming radial-basis kernel admits the same character decomposition used for IQP-QCBM training. Input-dependent rotations are established condition- encoding primitives in quantum circuit learning and feature-space models [15–17]. Circuit-Fourier theory describes the resulting condition response as a finite Fourier series [18–20]; the IQP active-generator expansion further filters which frequencies enter a selected observable [14]. We use these established results to explain the model, rather than presenting a new Fourier theorem.

Our contribution is therefore a conditional extension of the classically trained IQP-QCBM framework. The condition changes only the prepared product state, so one trained and compiled circuit represents an entire family $q_\theta(x \mid c)$. We first make that fixed-circuit requirement precise and show how the existing estimator and response descriptions carry over. We then test the construction on a controlled family of Gaussian Hamming-weight distributions. Across sizes and widths, the shared circuit reliably tracks a moving target and interpolates to unseen intermediate conditions relative to a matched circuit that receives no condition. Separate circuits trained for individual neighboring conditions remain a stronger fidelity reference, revealing the cost of sharing one circuit rather than obscuring it. We then challenge the same mechanism with eight-spin Curie–Weiss Gibbs laws. This family changes shape with inverse temperature, so it tests qualitative response rather than another translated peak. A complete unrestricted IQP circuit acts on eight visible qubits and up to nine latent ancillas; the visible output is their exact marginal. The deployed phases and compiled gates remain fixed across temperature. An ancilla-count ladder separates the headline 17-qubit model from the question of whether its improvement is merely a smooth response already available without latent qubits, while matched no-condition and single-temperature circuits provide progressively stronger references.

We use *conditional generalization* once in a deliberately narrow sense: interpolation to conditions unseen during training but strictly inside the training interval. We use *interpolation* hereafter. Extrapolation, broader distribution shift, universal conditional expressivity, and quantum advantage are outside scope.

## 2 Related work and scope of classical trainability

**Quantum circuit Born machines.** QCBMs are implicit generative models: a parameterized circuit defines a bitstring distribution by Born's rule and produces samples by measurement. Early work developed differentiable MMD training, hardware-oriented shallow circuit learning, and many-body constructions tied to sampling hardness [1–3]. Subsequent studies examined alternative divergences, the quantum-versus-classical learnability of discrete distributions, and generalization beyond memorizing observed bitstrings [4, 21–23]. This paper uses the same Born-machine semantics, but its question is not whether an arbitrary parameterized circuit can be trained from quantum samples. It asks whether one restricted IQP circuit can represent a condition-indexed family while retaining a classical training signal.

**Classically trained IQP Born machines.** Diagonal commuting circuits occupy a boundary between restricted classical simulation and sampling-hardness results [9, 12, 24]. The IQPopt and train-classical/deploy- quantum frameworks exploit probabilistic estimation of selected observables to optimize large IQP-QCBMs with MMD on classical hardware [8, 13, 14]. Recent analyses sharpen the scope of this promise. Fixed kernels and visible-only IQP models impose expressivity and trainability limits; kernel adaptation and ancillary qubits can enlarge the model class; and gradient scaling depends on topology, kernel spectrum, and initialization [25–28]. Spectral Born machines generalize the same classical-training principle through group Fourier structure [29]. We do not alter those optimization or sampling-complexity results. We extend their product-

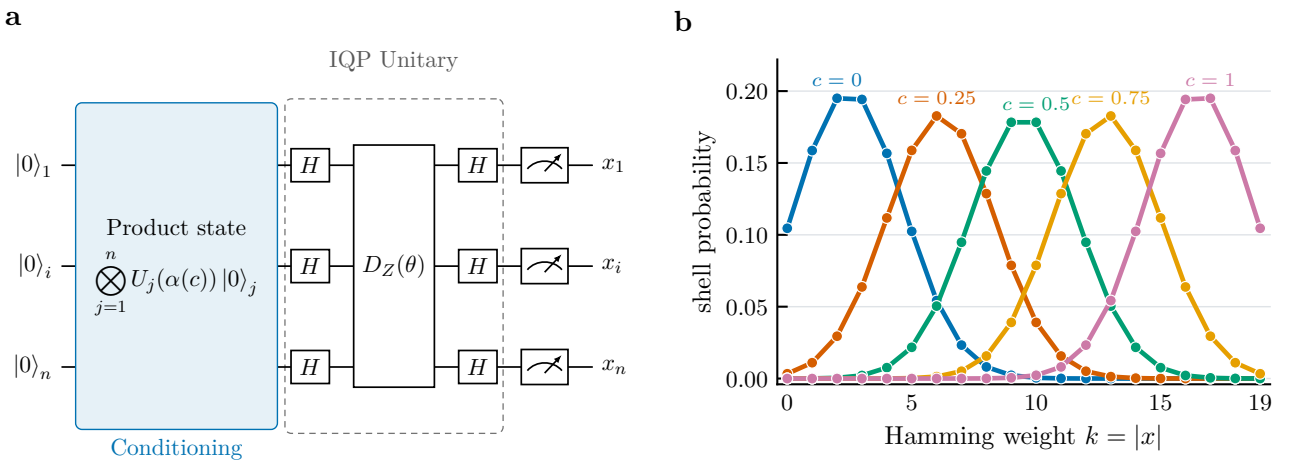



Figure 1: Product-state conditioning and the Gaussian target family. (a) Conditioning prepares the product input $\bigotimes_i U_i(\alpha(c))\,|0\rangle_i$; the IQP unitary and phase vector are shared across conditions. (b) Five representative 19-qubit target profiles; probability is uniform within each Hamming shell.

input interface to carry a condition and empirically characterize the resulting shared model.

**Conditional quantum generation.** Conditional Born machines have generated high-energy-physics events by reuploading an external parameter throughout a variational circuit [5]; that study also evaluates an unseen intermediate condition, so conditional interpolation itself is not our novelty. A separate hybrid construction encodes class labels in additional qubits and combines the circuit with a classical network [6]. More broadly, quantum generative protocols can upload classical randomness or latent variables into parameterized circuits [30]. Our setting differs in the deployment contract: after training, the ordered generator set, compiled gate sequence, and phase vector are identical for every condition; only a separable input preparation changes. This restriction makes the condition a data-loading interface rather than a condition-dependent program.

**Encoding, kernels, and terminology.** Parameterized quantum circuits embed classical inputs in quantum feature spaces [16, 17]; repeated injection, or data re-uploading, is an established way to enrich their input response [31]. Their response is a Fourier series whose frequencies are controlled by the encoding generators [18–20]. We use that language to describe *response across conditions*, not to claim universal function approximation. The training loss is an MMD two-sample discrepancy [32]. This is a classical positive-definite kernel on bitstrings, not a quantum feature-map kernel; its Hamming-character expansion is what connects the loss to estimable Pauli observables. Throughout the paper, *classically trainable* therefore means that the declared loss and phase gradients are classically estimable at finite budgets. It does not mean that the full bitstring law is efficiently available classically; full-probability simulation and classical output sampling remain outside the claim. Nor does the estimator guarantee an optimization landscape free of barren plateaus [33–36]. Likewise, our use of *interpolation* means evaluation at unseen condition values inside the training interval, whereas QCBM sample generalization concerns unseen bitstrings drawn from a fixed target law [23].

## 3 Product-state-conditioned IQP model

Let $g_j \in \{0,1\}^n$ identify a diagonal Pauli string $Z_{g_j} = \bigotimes_i Z_i^{g_{j,i}}$. For a fixed ordered generator bank $G = (g_1, \ldots, g_m)$ and phases $\theta \in \mathbb{R}^m$, define

$$D_Z(\theta) = \prod_{j=1}^{m} \exp(i\theta_j Z_{g_j}). \tag{1}$$

For a condition $c$, the output law is

$$q_\theta(x \mid c) = \left| \langle x|H^{\otimes n} D_Z(\theta) H^{\otimes n} \times \bigotimes_{i=1}^{n} R_Y(\alpha_i(c))|0\rangle \right|^2. \tag{2}$$

A model is *fixed-circuit conditional* when the compiled sequence, ordered $G$, and $\theta$ are identical for every condition; only the initial angles may vary. In the main campaign,

$$\alpha_i(c) = \pi(0.35 - 0.70c), \qquad i = 1, \ldots, n. \quad (3)$$

The ansatz contains all one- and two-qubit diagonal generators, hence $m = n(n+1)/2$ phases. The implementation hashes the generator matrix and checkpoint and verifies their invariance across all evaluated conditions.

For the Curie–Weiss experiment, eight measured qubits are augmented by $r$ ancillas and the complete circuit acts on all $8 + r$ qubits. The conditional visible law is

$$q_\theta(x \mid \beta) = \sum_{z\in\{0,1\}^r} \left| \langle x, z | H D_Z(\theta) H \times \bigotimes_{i=1}^{8+r} R_Y(\alpha(\beta)) \, |0\rangle \right|^2. \quad (4)$$

Here and below the tensor powers of $H$ are implicit. We broadcast the same angle to visible and ancillary inputs, apply unrestricted phases to every one- and two-body generator, and trace out the ancillas at measurement. Thus the $r = 9$ headline model has one fixed 17-qubit circuit with 153 phases; the ladder $r \in \{0, 1, 3, 5, 7, 9\}$ has respectively $36, 45, 66, 91, 120$, and $153$ phases. Changing $r$ changes both latent dimension and parameter count, so the ladder is a capacity ablation rather than a causal test of marginalization alone.

Two controls clarify what is gained by conditioning. The parameter-matched *no-condition* model receives the neutral input $c = 1/2$ for every target. The *per-condition reference* trains a separate circuit at each training condition. At a held-out coordinate, we compare against (i) a target-aware oracle that selects the more accurate of the two neighboring reference outputs and (ii) a target-blind barycentric mixture of those outputs. If $c_L < c < c_R$, the mixture is

$$q_{\text{mix}}(x \mid c) = \frac{c_R - c}{c_R - c_L} q_L(x) + \frac{c - c_L}{c_R - c_L} q_R(x). \quad (5)$$

Its weights use coordinates only; target metrics never enter. The adjacent-reference oracle is intentionally generous because it may inspect the held-out target to choose between $q_L$ and $q_R$.

## 4 Classical training signals

For the Hamming radial basis kernel $k_\sigma(x, y) = \exp[-d_H(x, y)/(2\sigma^2)]$, draw a mask $a \in \{0, 1\}^n$ with independent coordinates

$$a_i \sim \text{Bernoulli}(q_\sigma), \qquad q_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}. \quad (6)$$

Then

$$k_\sigma(x, y) = \mathbb{E}_a[(-1)^{a\cdot x}(-1)^{a\cdot y}]. \quad (7)$$

Thus squared MMD is an average of Pauli-character discrepancies. Independent model and reference draws yield the standard unbiased U-statistic [32]; differentiation of its finite sampled form gives phase gradients. We average bandwidths $(1, 2, 4)$ and conditions. Operator masks, model/reference samples, target samples, minibatches, and validation streams have separately derived seeds.

This establishes access to the declared training signal, not to every output probability. The Gaussian campaign samples 64 masks and 8,192 model/reference samples per bandwidth in training, then 256 masks and 16,384 samples in validation. The eight-visible-qubit Curie–Weiss campaign instead integrates all 256 visible masks with their exact kernel probabilities at every training and validation evaluation; each mask is padded by identity on the ancillas. It retains 8,192 model/reference samples in training and 16,384 in validation. Exact output probabilities are enumerated only after checkpoint selection. These controlled audits are feasible at the tested sizes, but are not an efficient asymptotic full-distribution procedure.

## 5 Established response structure

The following propositions collect established structure in the notation of Eq. (2); derivations appear in the supplement.

**Proposition 1** (Affine product-input support)**.** *For fixed $G, \theta, b, W$ and output $x$, if $\alpha(c) = b + Wc$, the support of $c \mapsto q_\theta(x \mid c)$ is contained in*

$$\Omega_{\text{full}}(W) = \left\{ \sum_{i=1}^{n} s_i w_i : s_i \in \{-1, 0, 1\} \right\}, \quad (8)$$

*where $w_i^\mathsf{T}$ is row $i$ of $W$.*

This is a direct specialization of multivariate circuit-Fourier theory [18–20]. The inclusion can be strict because frequencies coincide or coefficients cancel, and it gives no efficient procedure for evaluating the coefficients.

**Proposition 2** (Observable filter)**.** *For $F_a(c) = \langle Z_a \rangle_c$, only generators satisfying $a \cdot g_j = 1 \pmod 2$ enter the IQP active-generator expansion. Group their subsets by XOR mask $h$. A term survives on the real $R_Y$ product input only if $h \wedge a = 0$, and its condition frequencies are signed sums of rows $w_i$ with $i \in \mathrm{supp}(a) \cup \mathrm{supp}(h)$.*

This combines the established active-generator formula [14] with $\langle Y_i \rangle = 0$ on a real product state. Selected observable responses can therefore occupy a smaller space than full probabilities. At fixed encoding width both spaces are finite dimensional, which precludes a claim of universal density in arbitrary continuous conditional laws.

## 6 Gaussian benchmark and frozen protocol

The target is exchangeable within each Hamming shell. At system size $n$ and condition $c \in [0, 1]$, the requested mean is

$$\mu_n(c) = n(0.15 + 0.70c), \tag{9}$$

and the nominal width is $s\sqrt{n}$ for $s \in \{0.35, 0.50, 0.75\}$. We discretize a Gaussian over weights $0, \ldots, n$ and recalibrate its untruncated location numerically so the truncated distribution has mean $\mu_n(c)$. Its shell probability is then distributed uniformly over the $\binom{n}{k}$ strings at weight $k$. Profile TVD, the TVD between shell-mass vectors, is the primary metric; raw bitstring TVD, mean absolute error, width absolute error, and exact Conditional MMD are secondary diagnostics.

The factorial crosses $n \in \{8, 12, 16, 19\}$, three widths, and ten matched optimizer/data seeds. Shared conditioned and no-condition models train on $c \in \{0, 1/4, 1/2, 3/4, 1\}$; five separately trained circuits form the per-condition reference. All shared models are evaluated at those five coordinates and the common unseen grid $c \in \{1/16, 3/16, \ldots, 15/16\}$. No held-out samples are generated before exact post-selection evaluation.

Every model receives 2,000 Adam updates with cosine learning-rate decay $0.01 \to 0.001$. The total target minibatch contains 450 strings, balanced across conditions; an independent model also uses 450 samples per update. Validation occurs every 25 updates, and checkpoint selection uses only minimum validation Conditional MMD. Shared and no-condition models have matched initialization and estimator streams. Reference-model streams are tagged by training condition. The factorial contributes 840 trained models.

The condition-count study fixes $n = 19$ and $s = 0.50$. Nested grids with $K \in \{2, 3, 5, 9\}$ are evaluated under 10,000/2,000 training/validation draws per condition and under fixed totals of 18,000/3,600 divided among conditions. Each target has one deterministic 10,000/2,000 master pool; fixed-total datasets are prefixes. Nine separately trained per-condition circuits supply a deliberately generous capacity reference, with nested checkpoints reused only by declared identity. Together with the factorial, the study contains exactly 1,020 trained models.

A 100-update $n = 19, K = 9$ resource preflight recorded 22.02 seconds and 1.96 GB peak resident memory on the execution host, projecting 440.30 seconds per model. This passed the preregistered one-hour and 28 GiB resource limits. These are resource checks, not performance thresholds. Full manifest, dataset, stream, checkpoint, probability, environment, runtime, source, and revision digests are archived.

## 7 Conditional control and interpolation

We average conditions within each optimizer seed before method comparison and bootstrap the ten paired seed effects 10,000 times with seed 20260811. A strictly positive 95% interval is described as supported, a strictly negative interval as unsupported, and an interval containing zero as mixed. These are descriptive labels, not a publication gate. Every one of the 12 cells remains visible.

For the representative $n = 19, s = 0.50$ cell, no-condition minus conditioned training profile TVD is 0.2668 [0.2587, 0.2737]; held-out target-aware adjacent-reference minus conditioned profile TVD is 0.0016 [-0.0012, 0.0043]; and held-out coordinate-weighted mixture minus conditioned profile TVD is -0.0029 [-0.0053, -0.0006]. The conditioned-minus-per-condition-reference training fidelity gap is 0.0878 [0.0860, 0.0897] (pos-

itive values mean a cost of sharing). The corresponding raw-bitstring adjacent-reference effect is -0.0606 [-0.0749, -0.0461]. Across the full surface, conditioning improves the training-condition Hamming-weight profiles over the no-condition model in all 12 size–width settings. At unseen conditions, separately trained neighboring circuits are usually more accurate: their target-aware oracle is better in ten settings and statistically indistinguishable in two. The coordinate-weighted mixture is better in nine settings, indistinguishable in one, and worse in two. Thus the experiment shows robust conditional control and interpolation relative to no conditioning, together with a measurable fidelity cost for sharing one circuit across conditions.

## 8 Performance across size, width, and peak location

Figure 3 resolves the aggregate statements into the complete $4 \times 3$ performance surface. Positive and negative cells use the same color scale, and the numerical paired effect is printed in each cell. The per-condition curves and exhaustive intervals in the supplement reveal whether an aggregate seed mean masks a difficult peak location. In particular, the endpoints probe truncation asymmetry, while the common odd-sixteenth grid makes held-out comparisons identical across all training grids.

The width sweep changes two distinct difficulties. Narrow profiles demand accurate concentration in weight, while wider profiles allocate mass to more shells. Increasing $n$ also increases both the output space and the complete two-local parameter count. We therefore avoid pooling cells across $n$ or $s$ into one headline effect. The heatmaps summarize direction and magnitude; the table in the supplement retains the paired interval and cell-level wins.

## 9 Condition-count and data-allocation scaling

The $K$ study asks whether adding training conditions helps because the model sees the conditional curve more densely or merely because it receives more data. The fixed-per-condition regime increases total samples with $K$; the fixed-total regime holds total samples constant. Because all subsets are prefixes of the same target-specific master pools, differences cannot be attributed to unrelated draws. We report the per-condition and worst- condition profile TVD rather than only the overall mean.

The shared circuit always has 190 phases and 2,000 optimizer updates. An per-condition reference has $190K$ stored phases and $2,000K$ optimizer updates. Its larger optimization work is part of the baseline definition, not hidden from the comparison. The nine-circuit per-condition reference is particularly generous: its nested outputs are reused for all $K$ values, whereas shared circuits are retrained for their declared grid and data regime.

## 10 Temperature control in the Curie–Weiss model

The Gaussian task isolates conditional translation; Curie–Weiss tests whether the same fixed-circuit mechanism can follow a family that changes qualitatively with its condition. For spins $s_i \in \{-1, +1\}$ and $M = \sum_i s_i$ on eight visible sites, we use

$$H(s) = -\frac{M^2}{16}, \qquad p_\beta(s) = \frac{e^{-\beta H(s)}}{Z_\beta}. \tag{10}$$

At zero field the law is invariant under spin flip and permutation. Raising $\beta$ moves the finite-size magnetization profile from a disordered central distribution, through enhanced fluctuations near the thermodynamic critical value $\beta_c = 1$, toward an ordered bimodal distribution. Magnetization-profile TVD and raw bitstring TVD are co-primary. The first tests whether mass is assigned to the correct magnetization shells; the second also tests whether that mass is distributed uniformly within each shell.

We retain the Gaussian encoder without adapting it to Curie–Weiss outcomes: $c = \beta - 0.5$ and $\alpha(\beta) = \pi[0.35 - 0.70c]$. Six training temperatures $\{0.5, 0.7, 0.9, 1.1, 1.3, 1.5\}$ alternate with five unseen midpoints $\{0.6, 0.8, 1.0, 1.2, 1.4\}$. In particular, $\beta = 1$ is never available during optimization or checkpoint selection. The conditioned circuit broadcasts this angle to eight visible inputs and $r$ ancilla inputs, uses an unrestricted complete two-local phase vector, and marginalizes the ancillas. We train it and a matched no-condition control

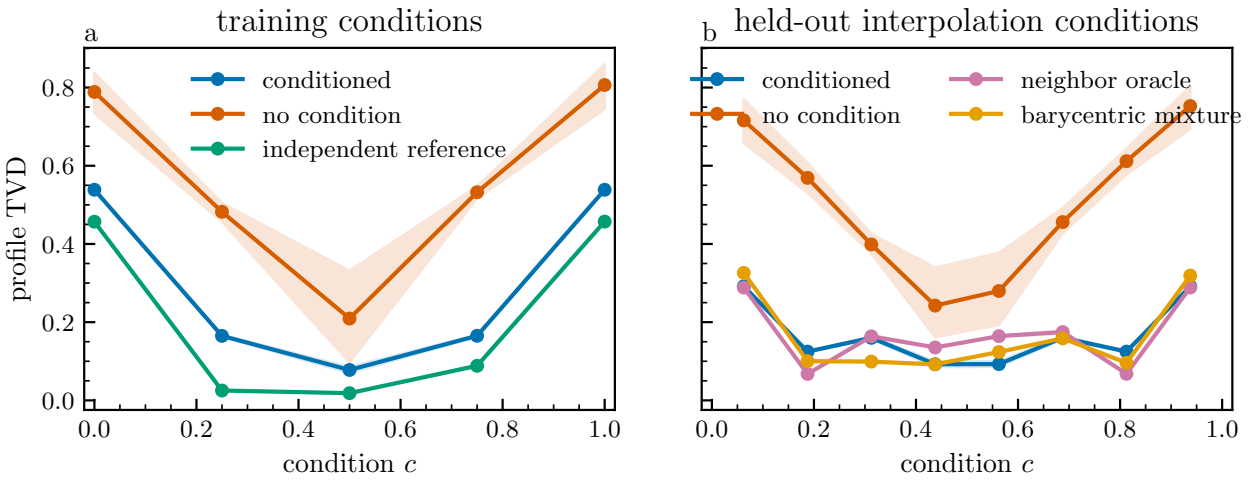



Figure 2: Training-condition control and held-out interpolation at $n = 19, s = 0.50$. Lines are ten-seed means and bands are deterministic 95% seed-bootstrap intervals. The held-out panel compares the conditioned circuit with no conditioning, a target-aware choice between the two adjacent per-condition circuits, and their coordinate-weighted mixture.

at $r \in \{0, 1, 3, 5, 7, 9\}$. The control receives the neutral $\beta = 1$ input for every target. At $r = 9$, six additional circuits are each trained for one temperature. Their adjacent pair defines a target-blind equal-mixture baseline at every held-out midpoint and two target-aware oracles, selected separately by profile and raw TVD.

Ten matched data and optimizer seeds give 60 conditioned fits, 60 no-condition fits, and 60 single-temperature references: 180 atomic fits with no restart selection. Each fit uses 2,000 Adam updates with cosine decay from 0.01 to 0.001, 75 examples per training temperature, and validation every 25 updates. The checkpoint is selected only by validation Conditional MMD. The exact visible law is evaluated at all eleven temperatures after selection. We average temperatures within seed and form 95% paired intervals from 10,000 bootstrap resamples. No empirical pass threshold is imposed.

The immutable ancilla Curie–Weiss campaign is not yet complete. We therefore report its frozen design but no performance direction, numerical interval, or thermal-state-preparation interpretation in this working draft.

## 11 Compact Gaussian ablations

Two archived studies test modeling choices outside the frozen factorial. First, complete two-local connectivity is compared with lattice and heavy-hex generator banks on a two-peak 19-qubit task. Second, the broadcast-angle map is compared with a symmetric-reupload encoding at matched parameter count. Their target grids and training protocols differ from the 1,020-model campaign, so both remain descriptive supplement evidence. The older Gaussian ancilla study is also retained only in the supplement and repository: its protocol is not commensurate with the Curie–Weiss ancilla ladder and it does not enter the main narrative. Target-symmetry-aware post-processing is likewise supplementary and excluded from every raw-circuit claim.

## 12 Limitations, cost, and reproducibility

Both target families are one-dimensional and exchangeable. Low profile TVD means that the model places the right total mass in each Hamming shell; it does not mean that probability is uniform among strings inside a shell. The unrestricted Curie–Weiss circuit does not build exchange symmetry into its phases, so profile and raw bitstring fidelity can differ substantially. We therefore make both TVDs primary for that study. The evidence concerns eight visible spins, one Curie–Weiss family, one fixed 17-qubit headline circuit, and interpolation inside the trained interval. It does not establish extrapolation, robustness to shifted Hamiltonians, or universal condi-

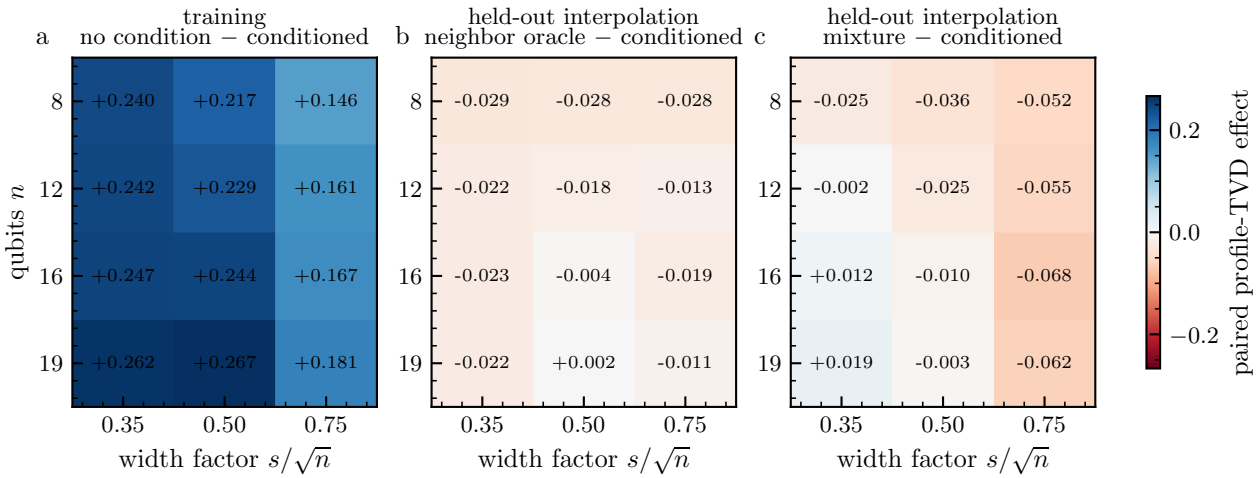



Figure 3: Complete Gaussian performance surface. Each entry is a ten-seed paired profile-TVD effect: no-condition minus conditioned at training conditions, and adjacent-reference oracle or coordinate-weighted mixture minus conditioned at held-out conditions. Positive values favor the conditioned circuit.

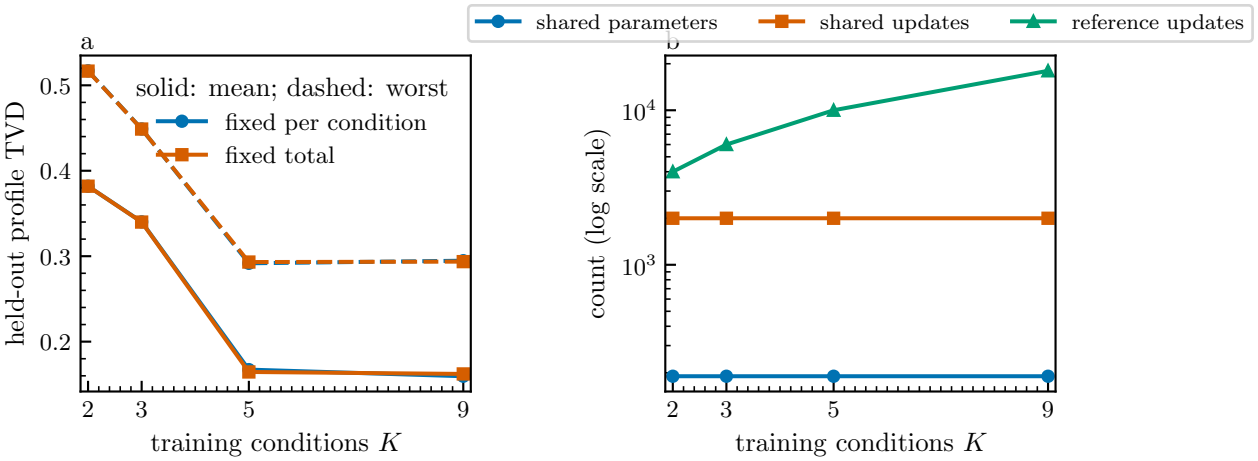



Figure 4: Condition-count characterization at $n = 19, s = 0.50$. (a) Mean and worst-condition held-out profile TVD for the fixed-per-condition and fixed-total regimes. (b) Shared-model parameter count and optimization work compared with independently trained references.

tional generation.

Our operational classical-trainability statement is likewise bounded. The training objective and gradients use finite classical estimators, but exact post-selection evaluation enumerates $2^n$ probabilities. Runtime grows with system size, connectivity, estimator budget, and optimizer work. Nothing here establishes efficient classical sampling of the trained output, a quantum advantage, or a favorable asymptotic optimization landscape.

Curie–Weiss is a controlled finite-size probability benchmark, not a claim of state-of-the-art thermal-state preparation or exact critical scaling. The isotropic Hamming kernel is aligned with magnetization shells but does not resolve spatial correlations. The ancilla ladder also confounds latent width with phase count: it has no matched-capacity higher-local visible control and cannot attribute a gain to marginalization alone. General many-body Hamiltonians, extrapolation in temperature, larger visible systems, and nonexchangeable physical observables remain outside the evidence.

Both campaigns are manifest-driven and isolated. The Gaussian store declares 1,020 identities; the Curie–Weiss ancilla store declares 180. They reject undeclared or duplicate completion, bind role-tagged datasets and estimator streams, and refuse partial-result reuse. Execution is

resumable and shardable. Each result binds the manifest, source tree, environment, revision, dataset, checkpoint, probability archive, invocation, runtime, and peak memory. Curie–Weiss figures, tables, and claims are generated only from its complete report. The supplement describes regeneration, archive checks, and clean-build verification.

## 13 Conclusion

Product-state conditioning gives a precise fixed-deployment route to conditional IQP generation: one compiled commuting circuit and one phase vector respond to a classical condition through local input rotations. The selected Hamming-kernel training objective and gradients remain accessible to classical probabilistic estimation, while full-output access is not claimed. Across the Gaussian campaign, conditioning improves training-condition profile TVD in all 12 size–width cells. Held-out outputs, however, usually do not outperform the generous adjacent-reference oracle or its target-blind barycentric mixture: the adjacent-reference comparison is mixed in two cells and unsupported in ten, while the mixture comparison is supported in two, mixed in one, and unsupported in nine. The result is therefore evidence for robust conditional control, not a broad claim of superior interpolation. This positive-and-negative performance map provides a transparent foundation for improving fixed-circuit conditional models while keeping the present claim restricted to the tested Gaussian Hamming-profile family. The Curie–Weiss extension remains evidence-gated until its complete immutable 180-fit campaign is finalized.

## Data and code availability

The source repository contains the frozen manifest, role-tagged datasets, immutable ledgers, checkpoints, exact probability archives, report, claim registry, figure/table generators, and clean-build scripts. The canonical PDF is promoted only after all declared identities and evaluations complete.

## Release metadata

Author names, affiliations, individual contributions, competing interests, arXiv category and license, and final human proof approval remain to be supplied and verified by the research team. This document is not upload-ready. No upload or submission is automated.

## References


[1] Jin-Guo Liu and Lei Wang. Differentiable learning of quantum circuit born machine. *Physical Review A*, 98(6):062324, 2018. DOI: 10.1103/PhysRevA.98.062324.

[2] Marcello Benedetti, Delfina Garcia-Pintos, Oscar Perdomo, Vicente Leyton-Ortega, Yunseong Nam, and Alejandro Perdomo-Ortiz. A generative modeling approach for benchmarking and training shallow quantum circuits. *npj Quantum Information*, 5:45, 2019. DOI: 10.1038/s41534-019-0157-8.

[3] Xun Gao, Zhengyu Zhang, and Luming Duan. A quantum machine learning algorithm based on generative models. *Science Advances*, 4(12): eaat9004, 2018. DOI: 10.1126/sciadv.aat9004.

[4] Chiara Leadbeater, Louis Sharrock, Brian Coyle, and Marcello Benedetti. $f$-divergences and cost function locality in generative modelling with quantum circuits. *Entropy*, 23(10):1281, 2021. DOI: 10.3390/e23101281.

[5] Oriel Kiss, Michele Grossi, Enrique Kajomovitz, and Sofia Vallecorsa. Conditional born machine for monte carlo event generation. *Physical Review A*, 106(2):022612, 2022. DOI: 10.1103/PhysRevA.106.022612.

[6] Qing-Wei Zeng, Hong-Ying Ge, Chen Gong, and Nan-Run Zhou. Conditional quantum circuit born machine based on a hybrid quantum–classical framework. *Physica A: Statistical Mechanics and its Applications*, 618:128693, 2023. DOI: 10.1016/j.physa.2023.128693.

[7] Dan Shepherd and Michael J. Bremner. Temporally unstructured quantum computation. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 465(2105): 1413–1439, 2009. DOI: 10.1098/rspa.2008.0443.

[8] Maarten Van den Nest. Simulating quantum computers with probabilistic methods. *Quantum Information and Computation*, 11(9–10): 784–812, 2011.

[9] Xiaotong Ni and Maarten Van den Nest. Commuting quantum circuits: Efficient classical simulations versus hardness results, 2012.

[10] Michael J. Bremner, Richard Jozsa, and Dan J. Shepherd. Classical simulation of commuting quantum computations implies collapse of the polynomial hierarchy. *Proceedings of the Royal Society A*, 467(2126):459–472, 2011. DOI: 10.1098/rspa.2010.0301.

[11] Michael J. Bremner, Ashley Montanaro, and Dan J. Shepherd. Average-case complexity versus approximate simulation of commuting quantum computations. *Physical Review Letters*, 117(8):080501, 2016. DOI: 10.1103/PhysRevLett.117.080501.

[12] Michael J. Bremner, Ashley Montanaro, and Dan J. Shepherd. Achieving quantum supremacy with sparse and noisy commuting quantum computations. *Quantum*, 1:8, 2017. DOI: 10.22331/q-2017-04-25-8.

[13] Erik Armengol and Joseph Bowles. IQPopt: Fast optimization of instantaneous quantum polynomial circuits in JAX, 2025.

[14] Erik Recio-Armengol, Shahnawaz Ahmed, and Joseph Bowles. Train on classical, deploy on quantum: Scaling generative quantum machine learning to a thousand qubits, 2025.

[15] Kosuke Mitarai, Makoto Negoro, Masahiro Kitagawa, and Keisuke Fujii. Quantum circuit learning. *Physical Review A*, 98(3):032309, 2018. DOI: 10.1103/PhysRevA.98.032309.

[16] Maria Schuld and Nathan Killoran. Quantum machine learning in feature hilbert spaces. *Physical Review Letters*, 122(4):040504, 2019. DOI: 10.1103/PhysRevLett.122.040504.

[17] Vojtěch Havlíček, Antonio D. Córcoles, Kristan Temme, Aram W. Harrow, Abhinav Kandala, Jerry M. Chow, and Jay M. Gambetta. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567(7747):209–212, 2019. DOI: 10.1038/s41586-019-0980-2.

[18] Francisco Javier Gil Vidal and Dirk Oliver Theis. Input redundancy for parameterized quantum circuits. *Frontiers in Physics*, 8:297, 2020. DOI: 10.3389/fphy.2020.00297.

[19] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. The effect of data encoding on the expressive power of variational quantum-machine-learning models. *Physical Review A*, 103(3):032430, 2021. DOI: 10.1103/PhysRevA.103.032430.

[20] Berta Casas and Alba Cervera-Lierta. Multidimensional fourier series with quantum circuits. *Physical Review A*, 107(6):062612, 2023. DOI: 10.1103/PhysRevA.107.062612.

[21] Ryan Sweke, Jean-Pierre Seifert, Dominik Hangleiter, and Jens Eisert. On the quantum versus classical learnability of discrete distributions. *Quantum*, 5:417, 2021. DOI: 10.22331/q-2021-03-23-417.

[22] Yuxuan Du, Zhuozhuo Tu, Bujiao Wu, Xiao Yuan, and Dacheng Tao. Power of quantum generative learning. *Physical Review Letters*, 131(14):140601, 2023. DOI: 10.1103/PhysRevLett.131.140601.

[23] Kaitlin Gili, Mohamed Hibat-Allah, Marta Mauri, Chris Ballance, and Alejandro Perdomo-Ortiz. Do quantum circuit born machines generalize? *Quantum Science and Technology*, 8(3): 035021, 2023. DOI: 10.1088/2058-9565/acd578.

[24] Yoshifumi Nakata and Mio Murao. Diagonal quantum circuits: Their computational power and applications. *European Physical Journal Plus*, 129(7):152, 2014. DOI: 10.1140/epjp/i2014-14152-9.

[25] Andrii Kurkin, Kevin Shen, Susanne Pielawa, Hao Wang, and Vedran Dunjko. Universality and kernel-adaptive training for classically trained, quantum-deployed generative models, 2025.

[26] Kevin Shen, Susanne Pielawa, Vedran Dunjko, and Hao Wang. Characterizing trainability of instantaneous quantum polynomial circuit born machines, 2026.

[27] Sacha Lerch, Joseph Bowles, Ricard Puig, Erik Armengol, Zo"e Holmes, and Supanut Thanasilp. IQP born machines under data-dependent and agnostic initialization strategies, 2026.

[28] Gennaro De Luca. Trainability of IQP quantum circuit born machines under gaussian initialization, 2026.

[29] Austin Huang, William Maxwell, Vasilis Belis, Evan Peters, Jason Pye, Soran Jahangiri, and Joseph Bowles. Spectral born machines: Classically trainable quantum generative models for discrete data, 2026.

[30] Oleksandr Kyriienko, Annie E. Paine, and Vincent E. Elfving. Protocols for trainable and differentiable quantum generative modeling. *Physical Review Research*, 6(3):033291, 2024. DOI: 10.1103/PhysRevResearch.6.033291.

[31] Adrián Pérez-Salinas, Alba Cervera-Lierta, Elies Gil-Fuster, and José I. Latorre. Data re-uploading for a universal quantum classifier. *Quantum*, 4:226, 2020. DOI: 10.22331/q-2020-02-06-226.

[32] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012. URL https://jmlr.org/papers/v13/gretton12a.html.

[33] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9:4812, 2018. DOI: 10.1038/s41467-018-07090-4.

[34] M. Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, 2021. DOI: 10.1038/s42254-021-00348-9.

[35] Supanut Thanasilp, Samson Wang, M. Cerezo, and Zoë Holmes. Exponential concentration in quantum kernel methods. *Nature Communications*, 15:5200, 2024. DOI: 10.1038/s41467-024-49287-w.

[36] Manuel S. Rudolph, Sacha Lerch, Supanut Thanasilp, Oriel Kiss, Oxana Shaya, Sofia Vallecorsa, Michele Grossi, and Zoë Holmes. Trainability barriers and opportunities in quantum generative modeling. *npj Quantum Information*, 10(1):116, 2024. DOI: 10.1038/s41534-024-00902-0.

# A Analytical details

This appendix derives the response and estimator statements used in the main text. They specialize established circuit-Fourier and probabilistic-IQP results to product-state-conditioned notation and are not presented as novel central theorems.

## A.1 Full-probability response support

One encoded qubit has density operator

$$\begin{aligned}\rho_i(\alpha_i) &= R_Y(\alpha_i)|0\rangle\langle 0|R_Y(\alpha_i)^\dagger \\ &= \tfrac{1}{2}\left(I + \sin\alpha_i X + \cos\alpha_i Z\right).\end{aligned} \tag{11}$$

Each entry is a linear combination of angle frequencies $-1, 0, +1$. Tensoring and substituting $\alpha_i(c) = b_i + w_i^\mathsf{T} c$ gives fixed matrices $A_s$ such that

$$\rho(c) = \sum_{s\in\{-1,0,1\}^n} A_s e^{i\sum_i s_i b_i} e^{i(\sum_i s_i w_i)^\mathsf{T} c}. \tag{12}$$

For fixed unitary $U$ and outcome $x$, $q(x \mid c) = \mathrm{Tr}[U^\dagger|x\rangle\langle x|U\rho(c)]$ is a fixed linear functional of $\rho(c)$ and introduces no new condition frequency. This proves Eq. (8). Zero rows, coincident sums, and coefficient cancellations can only shrink the support. The trace coefficient can still require exponential work to evaluate.

For the broadcast scalar encoding in Eq. (3), every row of $W$ is identical. Frequencies therefore collapse to integer multiples of the single slope, with multiplicities generated by different sign patterns. This degeneracy is useful for a one-coordinate location target but also limits the independent conditional directions available at fixed width.

## A.2 Pauli observable filter

The final Hadamards map measured $Z_a$ to $X_a$. A diagonal generator $Z_{g_j}$ satisfying $a\cdot g_j = 0 \pmod 2$ commutes with $X_a$ and cancels. For an active generator,

$$e^{-i\theta_j Z_{g_j}} X_a e^{i\theta_j Z_{g_j}} = X_a e^{2i\theta_j Z_{g_j}}. \tag{13}$$

Since the diagonal generators commute,

$$\begin{aligned}D_Z(\theta)^\dagger X_a D_Z(\theta) = X_a \prod_{j:a\cdot g_j=1} \big[&\cos(2\theta_j) I \\ &+ i\sin(2\theta_j) Z_{g_j}\big].\end{aligned} \tag{14}$$

Expanding groups active subsets by $h = \bigoplus_{j\in S} g_j$. Conjugating $X_a Z_h$ through the first Hadamards produces local $Z_a X_h$ factors. On the real product input,

$$\langle X_i\rangle = \sin\alpha_i, \qquad \langle Z_i\rangle = \cos\alpha_i, \qquad \langle Y_i\rangle = 0. \tag{15}$$

Every term with $h\wedge a \neq 0$ therefore vanishes. Surviving terms factor into sines and cosines on $\mathrm{supp}(a)\cup\mathrm{supp}(h)$; expanding into exponentials yields the signed sums stated in the proposition.

## A.3 Hamming-kernel mask identity and estimator

For one bit, let $a \sim \mathrm{Bernoulli}(q)$. Then

$$\mathbb{E}_a[(-1)^{a(x-y)}] = \begin{cases} 1, & x = y, \\ 1-2q, & x \neq y. \end{cases} \tag{16}$$

Choosing $q = q_\sigma$ from Eq. (6) gives $1-2q = e^{-1/(2\sigma^2)}$. Independence across bits yields Eq. (7). Let $\phi_a(x) = (-1)^{a\cdot x}$. For target samples $y_1,\ldots,y_T$ and model samples $x_1,\ldots,x_R$, the fixed-mask unbiased squared

discrepancy is

$$\widehat{\Delta}_a^2 = \frac{1}{R(R-1)} \sum_{r \neq r'} \phi_a(x_r)\phi_a(x_{r'}) + \frac{1}{T(T-1)} \sum_{t \neq t'} \phi_a(y_t)\phi_a(y_{t'}) - \frac{2}{RT} \sum_{r,t} \phi_a(x_r)\phi_a(y_t). \tag{17}$$

Averaging over masks, bandwidths, and conditions gives sampled Conditional MMD. Differentiation acts only on the model expectation because target and mask laws do not depend on phase parameters. This gives estimator access, not a lower bound on gradient magnitude or a convergence guarantee.

After checkpoint selection, the exact population value is computed from Walsh–Hadamard coefficients. With $\widehat{p}(a) = \sum_x p(x)(-1)^{a \cdot x}$,

$$\mathrm{MMD}_\sigma^2(p,q) = \sum_{a \in \{0,1\}^n} q_\sigma^{|a|}(1-q_\sigma)^{n-|a|} \left[\widehat{p}(a) - \widehat{q}(a)\right]^2. \tag{18}$$

## B Target construction and data isolation

For a nominal location $\ell$ and width $s\sqrt{n}$, define unnormalized shell mass

$$r_k(\ell, s) = \exp\left[-\frac{(k-\ell)^2}{2s^2 n}\right], \qquad k = 0, \ldots, n. \tag{19}$$

Truncation changes the realized mean, so a monotone scalar root solve chooses $\ell$ such that $\sum_k k r_k / \sum_k r_k = \mu_n(c)$. The resulting normalized shell mass $P(k \mid c)$ induces

$$p(x \mid c) = \frac{P(|x| \mid c)}{\binom{n}{|x|}}. \tag{20}$$

This is an exact analytic law. Training nonetheless sees only finite sampled strings.

Each combination of $n$, width factor, and master seed has one immutable data archive. It stores the nine-condition master grid, role-tagged train and validation strings, derived random seeds, and hashes. The main five-condition grid selects even master indices. The $K = 2, 3, 5$ grids are nested subsets of the nine-grid conditions. Fixed-total arrays are exact prefixes of the same 10,000/2,000 per-target masters. The archive schema has no held-out role, and the loader rejects any such role before training.

## C Curie–Weiss target, observables, and protocol

For a bitstring $x$, write $s_i = 2x_i - 1$, $M = \sum_i s_i$, and $m = M/n$. The exact target law is

$$p_\beta(x) = Z_\beta^{-1} \exp\left(\frac{\beta M^2}{2n}\right). \tag{21}$$

It depends only on Hamming weight, so all strings in one magnetization shell have equal probability. Spin flip maps $M$ to $-M$ and leaves the law unchanged. These identities are checked directly before any sample archive is written.

Exact post-selection evaluation records profile and raw bitstring TVD and

$$\chi = \beta n\big(\langle m^2 \rangle - \langle m \rangle^2\big), \tag{22}$$

$$U = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}, \tag{23}$$

$$C/n = \frac{\beta^2}{n} \mathrm{Var}(H), \tag{24}$$

together with $\langle |m| \rangle$, energy per spin, and exact multibandwidth MMD. At finite $n$, $\beta = 1$ is a crossover probe, not an estimate of an asymptotic critical exponent.

Table 1: Frozen data allocation in the condition-count study.

| $K$ | fixed per condition | | fixed total | |
|---|---|---|---|---|
| | train | validation | train | validation |
| 2 | 10,000 | 2,000 | 9,000 | 1,800 |
| 3 | 10,000 | 2,000 | 6,000 | 1,200 |
| 5 | 10,000 | 2,000 | 3,600 | 720 |
| 9 | 10,000 | 2,000 | 2,000 | 400 |

The new campaign reuses the ten immutable eight-qubit role-separated archives from the earlier study by declared path and SHA-256. Each contains 10,000 training and 2,000 validation strings at each of six training temperatures. Training and validation use independent hash-derived streams. The archive schema has neither held-out samples nor held-out seed fields. Exact laws at the five midpoints become accessible only after checkpoint selection.

For ancilla rung $r$, the complete two-local generator bank acts on $8 + r$ qubits. Every phase is optimized independently. Initial values are derived from optimizer seed and generator support, so every generator shared across rungs starts identically while newly available generators receive deterministic additional values. The angle $\alpha(\beta)$ is broadcast to all visible and ancilla inputs; every measured visible mask is padded by identity on the ancillas. Summing the exact joint law over all ancillary outcomes produces the archived 256-component visible law.

All 256 visible characters are integrated with their exact Hamming-kernel weights at each training and validation evaluation. This removes mask subsampling from the eight-visible-qubit comparison while retaining finite model/reference sampling. Conditioned and no-condition fits share initialization, minibatches, data, and estimator streams within each rung. Single-temperature references use the same seed-level nine-ancilla initialization with temperature-tagged data and streams.

Curie–Weiss ancilla result tables are withheld until all 180 result envelopes, checkpoints, probability archives, and report rows have matching digests.

## D Complete factorial results

Table 2 reports every size–width cell. Effects average conditions within seed and then pair by seed. Positive values favor the conditioned model. Status letters are S (95% interval strictly positive), M (contains zero), and U (strictly negative), ordered as training control, held-out adjacent-reference oracle, and held-out coordinate-weighted mixture. They describe the observed uncertainty and do not implement a success rule.

For auditability, the machine-readable report also records cell-level wins at each seed–condition pair, mean and width error, exact Conditional MMD, checkpoint selection, and model and target shell probabilities. The complete probability artifact retains all $2^n$ model probabilities at every declared evaluation coordinate.

## E Raw-bitstring diagnostics

Table 4 keeps the raw-law limitation adjacent to the primary profile result. Because targets are uniform within shells, profile TVD is a quotient metric: it ignores redistribution within a shell. The difference between profile and bitstring TVD quantifies that unmodeled structure. A positive paired effect still means the conditioned model is closer than the named baseline, but a large absolute TVD remains a limitation even when the relative comparison is favorable.

## F Condition-count results

Table 6 reports both data regimes, the average and worst held-out coordinate, parameter count, and optimization work. The shared-model parameter and update counts do not grow with $K$; the independent reference collection's stored parameters and optimizer work grow linearly. The per-condition JSON rows behind this table are retained so that a mean cannot hide a single difficult coordinate.

Table 8 exposes every held-out coordinate rather than leaving those values only in the machine-readable report. Entries are ten-seed means of profile TVD; lower is better.

Table 2: All factorial profile-TVD effects with deterministic paired 95% bootstrap intervals.

| $n$ | $s/\sqrt{n}$ | no condition − conditioned | adjacent-reference oracle − conditioned | mixture − conditioned | S/M/U |
|---|---|---|---|---|---|
| 8 | 0.35 | +0.2402 [+0.2278, +0.2536] | -0.0292 [-0.0468, -0.0108] | -0.0248 [-0.0418, -0.0064] | S/U/U |
| 8 | 0.50 | +0.2170 [+0.2015, +0.2418] | -0.0280 [-0.0336, -0.0225] | -0.0363 [-0.0432, -0.0292] | S/U/U |
| 8 | 0.75 | +0.1460 [+0.1293, +0.1714] | -0.0276 [-0.0339, -0.0214] | -0.0525 [-0.0584, -0.0461] | S/U/U |
| 12 | 0.35 | +0.2419 [+0.2352, +0.2505] | -0.0221 [-0.0255, -0.0190] | -0.0016 [-0.0052, +0.0019] | S/U/M |
| 12 | 0.50 | +0.2291 [+0.2213, +0.2391] | -0.0180 [-0.0213, -0.0146] | -0.0251 [-0.0283, -0.0216] | S/U/U |
| 12 | 0.75 | +0.1608 [+0.1515, +0.1722] | -0.0129 [-0.0151, -0.0107] | -0.0548 [-0.0567, -0.0526] | S/U/U |
| 16 | 0.35 | +0.2472 [+0.2421, +0.2519] | -0.0229 [-0.0279, -0.0174] | +0.0117 [+0.0067, +0.0172] | S/U/S |
| 16 | 0.50 | +0.2440 [+0.2354, +0.2537] | -0.0040 [-0.0071, +0.0001] | -0.0099 [-0.0133, -0.0057] | S/M/U |
| 16 | 0.75 | +0.1667 [+0.1570, +0.1770] | -0.0192 [-0.0205, -0.0178] | -0.0681 [-0.0696, -0.0666] | S/U/U |
| 19 | 0.35 | +0.2620 [+0.2594, +0.2645] | -0.0217 [-0.0240, -0.0192] | +0.0188 [+0.0164, +0.0213] | S/U/S |
| 19 | 0.50 | +0.2668 [+0.2587, +0.2737] | +0.0016 [-0.0012, +0.0043] | -0.0029 [-0.0053, -0.0006] | S/M/U |
| 19 | 0.75 | +0.1807 [+0.1648, +0.1999] | -0.0111 [-0.0165, -0.0049] | -0.0619 [-0.0670, -0.0560] | S/U/U |

Table 8: Per-condition held-out profile TVD in the condition-count study.

| $K$ | Regime | $c$ | Conditioned | No condition | Adjacent oracle | Mixture |
|---|---|---|---|---|---|---|
| 2 | fixed per condition | 0.0625 | 0.1687 | 0.5621 | 0.2885 | 0.3119 |
| 2 | fixed per condition | 0.1875 | 0.4635 | 0.6671 | 0.0677 | 0.1739 |
| 2 | fixed per condition | 0.3125 | 0.5170 | 0.7480 | 0.3494 | 0.4125 |
| 2 | fixed per condition | 0.4375 | 0.3785 | 0.7775 | 0.5992 | 0.5613 |
| 2 | fixed per condition | 0.5625 | 0.3784 | 0.7790 | 0.5991 | 0.5613 |
| 2 | fixed per condition | 0.6875 | 0.5169 | 0.7499 | 0.3492 | 0.4125 |
| 2 | fixed per condition | 0.8125 | 0.4634 | 0.6855 | 0.0679 | 0.1741 |
| 2 | fixed per condition | 0.9375 | 0.1687 | 0.5752 | 0.2886 | 0.3121 |
| 3 | fixed per condition | 0.0625 | 0.1816 | 0.5632 | 0.2885 | 0.3341 |
| 3 | fixed per condition | 0.1875 | 0.4140 | 0.5397 | 0.0677 | 0.2213 |
| 3 | fixed per condition | 0.3125 | 0.4473 | 0.5564 | 0.3494 | 0.2114 |
| 3 | fixed per condition | 0.4375 | 0.3163 | 0.5536 | 0.1354 | 0.1081 |
| 3 | fixed per condition | 0.5625 | 0.3171 | 0.5500 | 0.1643 | 0.1297 |
| 3 | fixed per condition | 0.6875 | 0.4489 | 0.5307 | 0.3492 | 0.2300 |
| 3 | fixed per condition | 0.8125 | 0.4150 | 0.5234 | 0.0679 | 0.2286 |
| 3 | fixed per condition | 0.9375 | 0.1813 | 0.5842 | 0.2886 | 0.3345 |
| 5 | fixed per condition | 0.0625 | 0.2917 | 0.7159 | 0.2885 | 0.3260 |
| 5 | fixed per condition | 0.1875 | 0.1247 | 0.5690 | 0.0677 | 0.1002 |
| 5 | fixed per condition | 0.3125 | 0.1599 | 0.3987 | 0.1638 | 0.0992 |
| 5 | fixed per condition | 0.4375 | 0.0925 | 0.2424 | 0.1354 | 0.0921 |
| 5 | fixed per condition | 0.5625 | 0.0927 | 0.2797 | 0.1643 | 0.1236 |
| 5 | fixed per condition | 0.6875 | 0.1601 | 0.4561 | 0.1747 | 0.1586 |
| 5 | fixed per condition | 0.8125 | 0.1248 | 0.6116 | 0.0679 | 0.0961 |
| 5 | fixed per condition | 0.9375 | 0.2916 | 0.7527 | 0.2886 | 0.3191 |
| 9 | fixed per condition | 0.0625 | 0.2947 | 0.7820 | 0.1439 | 0.2146 |

Continued on next page.

Table 3: Conditioned-model profile-TVD wins at the seed–Condition level in every factorial cell. Denominators are 50 Training comparisons and 80 Held-out comparisons per cell.

| $n$ | $s/\sqrt{n}$ | vs. no condition | vs. adjacent-reference oracle | vs. mixture |
|---|---|---|---|---|
| 8 | 0.35 | 50/50 | 23/80 | 28/80 |
| 8 | 0.50 | 50/50 | 8/80 | 10/80 |
| 8 | 0.75 | 50/50 | 19/80 | 6/80 |
| 12 | 0.35 | 48/50 | 30/80 | 39/80 |
| 12 | 0.50 | 48/50 | 18/80 | 21/80 |
| 12 | 0.75 | 50/50 | 20/80 | 20/80 |
| 16 | 0.35 | 46/50 | 11/80 | 50/80 |
| 16 | 0.50 | 48/50 | 38/80 | 30/80 |
| 16 | 0.75 | 50/50 | 20/80 | 20/80 |
| 19 | 0.35 | 46/50 | 27/80 | 58/80 |
| 19 | 0.50 | 47/50 | 40/80 | 38/80 |
| 19 | 0.75 | 49/50 | 28/80 | 20/80 |

Table 4: Raw bitstring TVD across all factorial cells. The middle columns are the conditioned model's absolute mean TVD; effects are paired 95% intervals.

| $n$ | $s/\sqrt{n}$ | training | held out | no condition − conditioned | adjacent-reference oracle − conditioned |
|---|---|---|---|---|---|
| 8 | 0.35 | 0.3394 | 0.2660 | +0.2388 [+0.2232, +0.2554] | -0.0243 [-0.0445, -0.0054] |
| 8 | 0.50 | 0.3086 | 0.2371 | +0.2055 [+0.1855, +0.2306] | -0.0380 [-0.0491, -0.0277] |
| 8 | 0.75 | 0.3229 | 0.2586 | +0.1288 [+0.1018, +0.1699] | -0.0099 [-0.0165, -0.0032] |
| 12 | 0.35 | 0.4179 | 0.3274 | +0.2578 [+0.2494, +0.2684] | -0.0371 [-0.0466, -0.0271] |
| 12 | 0.50 | 0.3661 | 0.2825 | +0.2594 [+0.2388, +0.2782] | -0.0356 [-0.0470, -0.0249] |
| 12 | 0.75 | 0.3596 | 0.2954 | +0.1687 [+0.1409, +0.1999] | +0.0156 [+0.0083, +0.0233] |
| 16 | 0.35 | 0.4618 | 0.3587 | +0.2515 [+0.2406, +0.2632] | -0.0429 [-0.0538, -0.0319] |
| 16 | 0.50 | 0.4061 | 0.3135 | +0.2575 [+0.2402, +0.2765] | -0.0418 [-0.0510, -0.0318] |
| 16 | 0.75 | 0.3929 | 0.3279 | +0.2100 [+0.1710, +0.2486] | +0.0250 [+0.0190, +0.0313] |
| 19 | 0.35 | 0.4964 | 0.3905 | +0.2394 [+0.2256, +0.2523] | -0.0610 [-0.0776, -0.0458] |
| 19 | 0.50 | 0.4380 | 0.3444 | +0.2559 [+0.2370, +0.2723] | -0.0606 [-0.0749, -0.0461] |
| 19 | 0.75 | 0.4292 | 0.3573 | +0.1940 [+0.1476, +0.2405] | +0.0074 [-0.0032, +0.0165] |

Table 5: Absolute secondary diagnostics for the conditioned model, averaged over Conditions and seeds. "Train" and "held" denote Training and Held-out coordinates.

| $n$ | $s/\sqrt{n}$ | mean absolute error train | mean absolute error held | width absolute error train | width absolute error held | exact Conditional MMD train | exact Conditional MMD held |
|---|---|---|---|---|---|---|---|
| 8 | 0.35 | 0.5797 | 0.2795 | 0.3714 | 0.3621 | 0.0184 | 0.0046 |
| 8 | 0.50 | 0.5964 | 0.3027 | 0.1428 | 0.1308 | 0.0193 | 0.0046 |
| 8 | 0.75 | 0.6319 | 0.3554 | 0.1332 | 0.2003 | 0.0208 | 0.0055 |
| 12 | 0.35 | 0.8565 | 0.3984 | 0.5225 | 0.5346 | 0.0204 | 0.0045 |
| 12 | 0.50 | 0.8969 | 0.4578 | 0.1463 | 0.1804 | 0.0220 | 0.0048 |
| 12 | 0.75 | 0.9641 | 0.5583 | 0.2829 | 0.3762 | 0.0244 | 0.0061 |
| 16 | 0.35 | 1.1360 | 0.5214 | 0.6114 | 0.6335 | 0.0206 | 0.0041 |
| 16 | 0.50 | 1.1909 | 0.6016 | 0.1426 | 0.2335 | 0.0228 | 0.0047 |
| 16 | 0.75 | 1.2908 | 0.7502 | 0.4486 | 0.5670 | 0.0259 | 0.0063 |
| 19 | 0.35 | 1.3490 | 0.6177 | 0.6631 | 0.6899 | 0.0204 | 0.0040 |
| 19 | 0.50 | 1.4049 | 0.7026 | 0.1503 | 0.2701 | 0.0228 | 0.0045 |
| 19 | 0.75 | 1.5071 | 0.8503 | 0.5448 | 0.6708 | 0.0263 | 0.0064 |

Table 6: Condition-count and data-allocation results at $n = 19, s = 0.50$.

| $K$ | regime | train/condition | val./condition | mean | worst | params | shared upd. | reference upd. |
|---|---|---|---|---|---|---|---|---|
| 2 | fixed per condition | 10,000 | 2,000 | 0.3819 | 0.5170 | 190 | 2,000 | 4,000 |
| 3 | fixed per condition | 10,000 | 2,000 | 0.3402 | 0.4489 | 190 | 2,000 | 6,000 |
| 5 | fixed per condition | 10,000 | 2,000 | 0.1673 | 0.2917 | 190 | 2,000 | 10,000 |
| 9 | fixed per condition | 10,000 | 2,000 | 0.1595 | 0.2947 | 190 | 2,000 | 18,000 |
| 2 | fixed total | 9,000 | 1,800 | 0.3819 | 0.5166 | 190 | 2,000 | 4,000 |
| 3 | fixed total | 6,000 | 1,200 | 0.3399 | 0.4488 | 190 | 2,000 | 6,000 |
| 5 | fixed total | 3,600 | 720 | 0.1645 | 0.2932 | 190 | 2,000 | 10,000 |
| 9 | fixed total | 2,000 | 400 | 0.1622 | 0.2936 | 190 | 2,000 | 18,000 |

| $K$ | Regime | $c$ | Conditioned | No condition | Adjacent oracle | Mixture |
|---|---|---|---|---|---|---|
| 9 | fixed per condition | 0.1875 | 0.1120 | 0.6182 | 0.1495 | 0.0472 |
| 9 | fixed per condition | 0.3125 | 0.1475 | 0.4024 | 0.1621 | 0.0684 |
| 9 | fixed per condition | 0.4375 | 0.0835 | 0.1596 | 0.1354 | 0.0621 |
| 9 | fixed per condition | 0.5625 | 0.0836 | 0.2007 | 0.1507 | 0.0725 |
| 9 | fixed per condition | 0.6875 | 0.1478 | 0.4499 | 0.1538 | 0.1058 |
| 9 | fixed per condition | 0.8125 | 0.1123 | 0.6518 | 0.1417 | 0.1452 |
| 9 | fixed per condition | 0.9375 | 0.2946 | 0.8084 | 0.1774 | 0.1890 |
| 2 | fixed total | 0.0625 | 0.1682 | 0.5652 | 0.2885 | 0.3119 |
| 2 | fixed total | 0.1875 | 0.4616 | 0.6572 | 0.0677 | 0.1739 |
| 2 | fixed total | 0.3125 | 0.5166 | 0.7354 | 0.3494 | 0.4125 |
| 2 | fixed total | 0.4375 | 0.3815 | 0.7669 | 0.5992 | 0.5613 |
| 2 | fixed total | 0.5625 | 0.3814 | 0.7618 | 0.5991 | 0.5613 |
| 2 | fixed total | 0.6875 | 0.5163 | 0.7272 | 0.3492 | 0.4125 |
| 2 | fixed total | 0.8125 | 0.4614 | 0.6647 | 0.0679 | 0.1741 |
| 2 | fixed total | 0.9375 | 0.1682 | 0.5744 | 0.2886 | 0.3121 |
| 3 | fixed total | 0.0625 | 0.1819 | 0.5630 | 0.2885 | 0.3341 |
| 3 | fixed total | 0.1875 | 0.4130 | 0.5417 | 0.0677 | 0.2213 |
| 3 | fixed total | 0.3125 | 0.4462 | 0.5575 | 0.3494 | 0.2114 |
| 3 | fixed total | 0.4375 | 0.3159 | 0.5515 | 0.1354 | 0.1081 |
| 3 | fixed total | 0.5625 | 0.3173 | 0.5480 | 0.1643 | 0.1297 |
| 3 | fixed total | 0.6875 | 0.4488 | 0.5325 | 0.3492 | 0.2300 |
| 3 | fixed total | 0.8125 | 0.4152 | 0.5262 | 0.0679 | 0.2286 |
| 3 | fixed total | 0.9375 | 0.1811 | 0.5862 | 0.2886 | 0.3345 |
| 5 | fixed total | 0.0625 | 0.2932 | 0.7198 | 0.2885 | 0.3260 |
| 5 | fixed total | 0.1875 | 0.1214 | 0.5725 | 0.0677 | 0.1002 |
| 5 | fixed total | 0.3125 | 0.1552 | 0.3974 | 0.1638 | 0.0992 |

Continued on next page.

Table 7: Condition-count paired profile-TVD effects with 95% intervals. Positive control, adjacent-reference-oracle, and mixture effects favor conditioning; a positive reference gap means the independent circuits have lower Training profile TVD.

| $K$ | Regime | no condition − conditioned | | condi- conditioned − reference | | adjacent-reference oracle − conditioned | | mixture − conditioned | |
|---|---|---|---|---|---|---|---|---|---|
| 2 | fixed per condition | +0.0465 | [+0.0289, +0.0250 | [+0.0221, -0.0557 | [-0.0697, -0.0417] | -0.0169 | [-0.0309, -0.0029] | | |
| | | +0.0682] | +0.0274] | | | | | | |
| 3 | fixed per condition | +0.1754 | [+0.1712, +0.1140 | [+0.1083, -0.1263 | [-0.1386, -0.1117] | -0.1155 | [-0.1278, -0.1008] | | |
| | | +0.1797] | +0.1194] | | | | | | |
| 5 | fixed per condition | +0.2668 | [+0.2587, +0.0878 | [+0.0860, +0.0016 | | [-0.0012, -0.0029 | [-0.0053, -0.0006] | | |
| | | +0.2737] | +0.0897] | | +0.0043] | | | | |
| 9 | fixed per condition | +0.3354 | [+0.3331, +0.0464 | [+0.0428, -0.0077 | [-0.0113, -0.0028] | -0.0464 | [-0.0502, -0.0412] | | |
| | | +0.3379] | +0.0497] | | | | | | |
| 2 | fixed total | +0.0524 | [+0.0339, +0.0251 | [+0.0226, -0.0557 | [-0.0693, -0.0424] | -0.0170 | [-0.0305, -0.0036] | | |
| | | +0.0724] | +0.0273] | | | | | | |
| 3 | fixed total | +0.1756 | [+0.1710, +0.1140 | [+0.1087, -0.1261 | [-0.1378, -0.1121] | -0.1152 | [-0.1270, -0.1013] | | |
| | | +0.1799] | +0.1192] | | | | | | |
| 5 | fixed total | +0.2693 | [+0.2620, +0.0859 | [+0.0823, +0.0043 | | [+0.0001, -0.0001 | [-0.0040, +0.0042] | | |
| | | +0.2739] | +0.0892] | | +0.0091] | | | | |
| 9 | fixed total | +0.3333 | [+0.3307, +0.0482 | [+0.0455, -0.0104 | [-0.0132, -0.0076] | -0.0491 | [-0.0521, -0.0463] | | |
| | | +0.3358] | +0.0510] | | | | | | |

| $K$ | Regime | $c$ | Conditioned | No condition | Adjacent oracle | Mixture |
|---|---|---|---|---|---|---|
| 5 | fixed total | 0.4375 | 0.0876 | 0.2355 | 0.1354 | 0.0921 |
| 5 | fixed total | 0.5625 | 0.0876 | 0.2752 | 0.1643 | 0.1236 |
| 5 | fixed total | 0.6875 | 0.1562 | 0.4579 | 0.1747 | 0.1586 |
| 5 | fixed total | 0.8125 | 0.1222 | 0.6127 | 0.0679 | 0.0961 |
| 5 | fixed total | 0.9375 | 0.2926 | 0.7549 | 0.2886 | 0.3191 |
| 9 | fixed total | 0.0625 | 0.2936 | 0.7815 | 0.1439 | 0.2146 |
| 9 | fixed total | 0.1875 | 0.1150 | 0.6173 | 0.1495 | 0.0472 |
| 9 | fixed total | 0.3125 | 0.1519 | 0.4010 | 0.1621 | 0.0684 |
| 9 | fixed total | 0.4375 | 0.0872 | 0.1583 | 0.1354 | 0.0621 |
| 9 | fixed total | 0.5625 | 0.0878 | 0.2006 | 0.1507 | 0.0725 |
| 9 | fixed total | 0.6875 | 0.1532 | 0.4498 | 0.1538 | 0.1058 |
| 9 | fixed total | 0.8125 | 0.1157 | 0.6518 | 0.1417 | 0.1452 |
| 9 | fixed total | 0.9375 | 0.2933 | 0.8087 | 0.1774 | 0.1890 |

# G Exhaustive paired intervals

The following machine-generated table lists every diagnostic comparison, not only the profile-TVD effects selected for the main text. The replication unit is always the matched optimizer seed; conditions are averaged before resampling.

Table 9: Paired intervals for $n = 8$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0243 [-0.0445, -0.0054] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | -0.0005 [-0.0006, -0.0004] | unsupported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.1103 [+0.0770, +0.1392] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0292 [-0.0468, -0.0108] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.0962 [-0.1592, -0.0276] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0328 [-0.0540, -0.0126] | unsupported |
| heldout mixture minus conditioned conditional mmd | -0.0006 [-0.0008, -0.0005] | unsupported |
| heldout mixture minus conditioned mean absolute error | -0.0030 [-0.0251, +0.0171] | mixed |
| heldout mixture minus conditioned profile tvd | -0.0248 [-0.0418, -0.0064] | unsupported |
| heldout mixture minus conditioned width absolute error | -0.0194 [-0.0820, +0.0489] | mixed |
| heldout no condition minus conditioned bitstring tvd | +0.2735 [+0.2572, +0.2906] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0406 [+0.0405, +0.0407] | supported |
| heldout no condition minus conditioned mean absolute error | +1.1205 [+1.0980, +1.1410] | supported |
| heldout no condition minus conditioned profile tvd | +0.2840 [+0.2695, +0.2990] | supported |
| heldout no condition minus conditioned width absolute error | +0.7802 [+0.6850, +0.8592] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0662 [+0.0474, +0.0856] | supported |
| training conditioned minus independent reference conditional mmd | +0.0074 [+0.0073, +0.0076] | supported |
| training conditioned minus independent reference mean absolute error | +0.1592 [+0.1460, +0.1735] | supported |
| training conditioned minus independent reference profile tvd | +0.0923 [+0.0766, +0.1065] | supported |
| training conditioned minus independent reference width absolute error | +0.0722 [+0.0179, +0.1220] | supported |
| training no condition minus conditioned bitstring tvd | +0.2388 [+0.2232, +0.2554] | supported |
| training no condition minus conditioned conditional mmd | +0.0525 [+0.0520, +0.0530] | supported |
| training no condition minus conditioned mean absolute error | +1.1017 [+1.0862, +1.1164] | supported |
| training no condition minus conditioned profile tvd | +0.2402 [+0.2278, +0.2536] | supported |
| training no condition minus conditioned width absolute error | +0.7978 [+0.7088, +0.8691] | supported |

Table 10: Paired intervals for $n = 8$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0380 [-0.0491, -0.0277] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | -0.0005 [-0.0006, -0.0003] | unsupported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.0431 [+0.0287, +0.0570] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0280 [-0.0336, -0.0225] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.0796 [-0.0909, -0.0692] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0362 [-0.0473, -0.0263] | unsupported |
| heldout mixture minus conditioned conditional mmd | -0.0003 [-0.0005, -0.0002] | unsupported |
| heldout mixture minus conditioned mean absolute error | -0.0377 [-0.0500, -0.0250] | unsupported |
| heldout mixture minus conditioned profile tvd | -0.0363 [-0.0432, -0.0292] | unsupported |
| heldout mixture minus conditioned width absolute error | -0.0028 [-0.0168, +0.0109] | mixed |
| heldout no condition minus conditioned bitstring tvd | +0.2375 [+0.2166, +0.2641] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0407 [+0.0405, +0.0409] | supported |
| heldout no condition minus conditioned mean absolute error | +1.0973 [+1.0828, +1.1116] | supported |
| heldout no condition minus conditioned profile tvd | +0.2698 [+0.2542, +0.2965] | supported |
| heldout no condition minus conditioned width absolute error | +0.8385 [+0.7480, +0.8905] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0411 [+0.0310, +0.0525] | supported |
| training conditioned minus independent reference conditional mmd | +0.0070 [+0.0069, +0.0071] | supported |
| training conditioned minus independent reference mean absolute error | +0.1843 [+0.1766, +0.1915] | supported |
| training conditioned minus independent reference profile tvd | +0.0788 [+0.0718, +0.0857] | supported |
| training conditioned minus independent reference width absolute error | +0.0366 [+0.0221, +0.0507] | supported |
| training no condition minus conditioned bitstring tvd | +0.2055 [+0.1855, +0.2306] | supported |
| training no condition minus conditioned conditional mmd | +0.0514 [+0.0509, +0.0522] | supported |
| training no condition minus conditioned mean absolute error | +1.0859 [+1.0750, +1.0977] | supported |
| training no condition minus conditioned profile tvd | +0.2170 [+0.2015, +0.2418] | supported |
| training no condition minus conditioned width absolute error | +0.8941 [+0.7986, +0.9503] | supported |

Table 11: Paired intervals for $n = 8$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0099 [-0.0165, -0.0032] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | +0.0015 [+0.0014, +0.0016] | supported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.1133 [+0.1009, +0.1227] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0276 [-0.0339, -0.0214] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.0902 [-0.1037, -0.0750] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0596 [-0.0653, -0.0544] | unsupported |
| heldout mixture minus conditioned conditional mmd | -0.0005 [-0.0006, -0.0005] | unsupported |
| heldout mixture minus conditioned mean absolute error | -0.1080 [-0.1201, -0.0993] | unsupported |
| heldout mixture minus conditioned profile tvd | -0.0525 [-0.0584, -0.0461] | unsupported |
| heldout mixture minus conditioned width absolute error | -0.1657 [-0.1743, -0.1574] | unsupported |
| heldout no condition minus conditioned bitstring tvd | +0.1557 [+0.1278, +0.1969] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0398 [+0.0395, +0.0404] | supported |
| heldout no condition minus conditioned mean absolute error | +1.0446 [+1.0327, +1.0538] | supported |
| heldout no condition minus conditioned profile tvd | +0.1900 [+0.1717, +0.2167] | supported |
| heldout no condition minus conditioned width absolute error | +0.5997 [+0.5068, +0.6518] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0503 [+0.0447, +0.0560] | supported |
| training conditioned minus independent reference conditional mmd | +0.0067 [+0.0066, +0.0068] | supported |
| training conditioned minus independent reference mean absolute error | +0.2277 [+0.2216, +0.2352] | supported |
| training conditioned minus independent reference profile tvd | +0.0688 [+0.0635, +0.0742] | supported |
| training conditioned minus independent reference width absolute error | +0.0995 [+0.0901, +0.1085] | supported |
| training no condition minus conditioned bitstring tvd | +0.1288 [+0.1018, +0.1699] | supported |
| training no condition minus conditioned conditional mmd | +0.0495 [+0.0490, +0.0506] | supported |
| training no condition minus conditioned mean absolute error | +1.0511 [+1.0412, +1.0592] | supported |
| training no condition minus conditioned profile tvd | +0.1460 [+0.1293, +0.1714] | supported |
| training no condition minus conditioned width absolute error | +0.7865 [+0.6961, +0.8358] | supported |

Table 12: Paired intervals for $n = 12$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0371 [-0.0466, -0.0271] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | -0.0004 [-0.0005, -0.0003] | unsupported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.0992 [+0.0886, +0.1100] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0221 [-0.0255, -0.0190] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.1215 [-0.1299, -0.1129] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0153 [-0.0247, -0.0051] | unsupported |
| heldout mixture minus conditioned conditional mmd | +0.0000 [-0.0001, +0.0001] | mixed |
| heldout mixture minus conditioned mean absolute error | +0.0297 [+0.0162, +0.0431] | supported |
| heldout mixture minus conditioned profile tvd | -0.0016 [-0.0052, +0.0019] | mixed |
| heldout mixture minus conditioned width absolute error | +0.0261 [+0.0165, +0.0359] | supported |
| heldout no condition minus conditioned bitstring tvd | +0.3019 [+0.2894, +0.3164] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0410 [+0.0409, +0.0411] | supported |
| heldout no condition minus conditioned mean absolute error | +1.7016 [+1.6905, +1.7132] | supported |
| heldout no condition minus conditioned profile tvd | +0.2892 [+0.2798, +0.2998] | supported |
| heldout no condition minus conditioned width absolute error | +0.5399 [+0.3070, +0.7984] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0509 [+0.0423, +0.0587] | supported |
| training conditioned minus independent reference conditional mmd | +0.0074 [+0.0073, +0.0075] | supported |
| training conditioned minus independent reference mean absolute error | +0.2118 [+0.2023, +0.2218] | supported |
| training conditioned minus independent reference profile tvd | +0.0831 [+0.0791, +0.0872] | supported |
| training conditioned minus independent reference width absolute error | +0.0752 [+0.0675, +0.0830] | supported |
| training no condition minus conditioned bitstring tvd | +0.2578 [+0.2494, +0.2684] | supported |
| training no condition minus conditioned conditional mmd | +0.0552 [+0.0541, +0.0562] | supported |
| training no condition minus conditioned mean absolute error | +1.6805 [+1.6708, +1.6909] | supported |
| training no condition minus conditioned profile tvd | +0.2419 [+0.2352, +0.2505] | supported |
| training no condition minus conditioned width absolute error | +0.5786 [+0.3458, +0.8372] | supported |

Table 13: Paired intervals for $n = 12$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0356 [-0.0470, -0.0249] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | -0.0001 [-0.0002, -0.0000] | unsupported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.0568 [+0.0496, +0.0653] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0180 [-0.0213, -0.0146] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.1159 [-0.1234, -0.1074] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0215 [-0.0328, -0.0107] | unsupported |
| heldout mixture minus conditioned conditional mmd | +0.0001 [+0.0000, +0.0002] | supported |
| heldout mixture minus conditioned mean absolute error | -0.0563 [-0.0635, -0.0469] | unsupported |
| heldout mixture minus conditioned profile tvd | -0.0251 [-0.0283, -0.0216] | unsupported |
| heldout mixture minus conditioned width absolute error | +0.0116 [+0.0043, +0.0190] | supported |
| heldout no condition minus conditioned bitstring tvd | +0.3024 [+0.2842, +0.3197] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0419 [+0.0415, +0.0424] | supported |
| heldout no condition minus conditioned mean absolute error | +1.6422 [+1.6358, +1.6508] | supported |
| heldout no condition minus conditioned profile tvd | +0.2793 [+0.2663, +0.2923] | supported |
| heldout no condition minus conditioned width absolute error | +0.8925 [+0.5774, +1.1911] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0312 [+0.0220, +0.0405] | supported |
| training conditioned minus independent reference conditional mmd | +0.0071 [+0.0070, +0.0072] | supported |
| training conditioned minus independent reference mean absolute error | +0.2773 [+0.2705, +0.2830] | supported |
| training conditioned minus independent reference profile tvd | +0.0877 [+0.0839, +0.0911] | supported |
| training conditioned minus independent reference width absolute error | +0.0343 [+0.0266, +0.0403] | supported |
| training no condition minus conditioned bitstring tvd | +0.2594 [+0.2388, +0.2782] | supported |
| training no condition minus conditioned conditional mmd | +0.0542 [+0.0524, +0.0560] | supported |
| training no condition minus conditioned mean absolute error | +1.6407 [+1.6322, +1.6518] | supported |
| training no condition minus conditioned profile tvd | +0.2291 [+0.2213, +0.2391] | supported |
| training no condition minus conditioned width absolute error | +1.0030 [+0.6903, +1.2993] | supported |

Table 14: Paired intervals for $n = 12$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | +0.0156 [+0.0083, +0.0233] | supported |
| heldout neighbor oracle minus conditioned conditional mmd | +0.0018 [+0.0014, +0.0023] | supported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.1243 [+0.0983, +0.1512] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0129 [-0.0151, -0.0107] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.0644 [-0.0878, -0.0402] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0476 [-0.0501, -0.0453] | unsupported |
| heldout mixture minus conditioned conditional mmd | -0.0001 [-0.0001, -0.0000] | unsupported |
| heldout mixture minus conditioned mean absolute error | -0.1791 [-0.1851, -0.1731] | unsupported |
| heldout mixture minus conditioned profile tvd | -0.0548 [-0.0567, -0.0526] | unsupported |
| heldout mixture minus conditioned width absolute error | -0.2365 [-0.2563, -0.2149] | unsupported |
| heldout no condition minus conditioned bitstring tvd | +0.1953 [+0.1689, +0.2248] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0408 [+0.0398, +0.0420] | supported |
| heldout no condition minus conditioned mean absolute error | +1.5417 [+1.5374, +1.5463] | supported |
| heldout no condition minus conditioned profile tvd | +0.2019 [+0.1903, +0.2138] | supported |
| heldout no condition minus conditioned width absolute error | +0.8031 [+0.5245, +1.0192] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0288 [+0.0256, +0.0321] | supported |
| training conditioned minus independent reference conditional mmd | +0.0066 [+0.0066, +0.0066] | supported |
| training conditioned minus independent reference mean absolute error | +0.3499 [+0.3473, +0.3527] | supported |
| training conditioned minus independent reference profile tvd | +0.0582 [+0.0554, +0.0613] | supported |
| training conditioned minus independent reference width absolute error | +0.0797 [+0.0647, +0.0938] | supported |
| training no condition minus conditioned bitstring tvd | +0.1687 [+0.1409, +0.1999] | supported |
| training no condition minus conditioned conditional mmd | +0.0500 [+0.0480, +0.0524] | supported |
| training no condition minus conditioned mean absolute error | +1.5664 [+1.5605, +1.5728] | supported |
| training no condition minus conditioned profile tvd | +0.1608 [+0.1515, +0.1722] | supported |
| training no condition minus conditioned width absolute error | +1.0492 [+0.7734, +1.2647] | supported |

Table 15: Paired intervals for $n = 16$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0429 [-0.0538, -0.0319] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | -0.0001 [-0.0001, +0.0000] | mixed |
| heldout neighbor oracle minus conditioned mean absolute error | +0.1446 [+0.1381, +0.1499] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0229 [-0.0279, -0.0174] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.1356 [-0.1557, -0.1146] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0060 [-0.0166, +0.0048] | mixed |
| heldout mixture minus conditioned conditional mmd | +0.0004 [+0.0003, +0.0005] | supported |
| heldout mixture minus conditioned mean absolute error | +0.0464 [+0.0369, +0.0546] | supported |
| heldout mixture minus conditioned profile tvd | +0.0117 [+0.0067, +0.0172] | supported |
| heldout mixture minus conditioned width absolute error | +0.0987 [+0.0783, +0.1208] | supported |
| heldout no condition minus conditioned bitstring tvd | +0.3087 [+0.2927, +0.3256] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0399 [+0.0397, +0.0400] | supported |
| heldout no condition minus conditioned mean absolute error | +2.2786 [+2.2711, +2.2848] | supported |
| heldout no condition minus conditioned profile tvd | +0.3139 [+0.3059, +0.3231] | supported |
| heldout no condition minus conditioned width absolute error | +0.6858 [+0.2739, +1.1408] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0464 [+0.0370, +0.0554] | supported |
| training conditioned minus independent reference conditional mmd | +0.0068 [+0.0067, +0.0069] | supported |
| training conditioned minus independent reference mean absolute error | +0.2799 [+0.2737, +0.2868] | supported |
| training conditioned minus independent reference profile tvd | +0.0835 [+0.0785, +0.0878] | supported |
| training conditioned minus independent reference width absolute error | +0.0675 [+0.0499, +0.0838] | supported |
| training no condition minus conditioned bitstring tvd | +0.2515 [+0.2406, +0.2632] | supported |
| training no condition minus conditioned conditional mmd | +0.0538 [+0.0525, +0.0550] | supported |
| training no condition minus conditioned mean absolute error | +2.2557 [+2.2469, +2.2646] | supported |
| training no condition minus conditioned profile tvd | +0.2472 [+0.2421, +0.2519] | supported |
| training no condition minus conditioned width absolute error | +0.7318 [+0.3236, +1.1828] | supported |

Table 16: Paired intervals for $n = 16$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0418 [-0.0510, -0.0318] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | +0.0001 [+0.0001, +0.0002] | supported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.0876 [+0.0800, +0.0953] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0040 [-0.0071, +0.0001] | mixed |
| heldout neighbor oracle minus conditioned width absolute error | -0.1386 [-0.1487, -0.1256] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0193 [-0.0282, -0.0093] | unsupported |
| heldout mixture minus conditioned conditional mmd | +0.0004 [+0.0003, +0.0005] | supported |
| heldout mixture minus conditioned mean absolute error | -0.0697 [-0.0805, -0.0566] | unsupported |
| heldout mixture minus conditioned profile tvd | -0.0099 [-0.0133, -0.0057] | unsupported |
| heldout mixture minus conditioned width absolute error | +0.0298 [+0.0163, +0.0472] | supported |
| heldout no condition minus conditioned bitstring tvd | +0.3069 [+0.2923, +0.3265] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0408 [+0.0403, +0.0413] | supported |
| heldout no condition minus conditioned mean absolute error | +2.1984 [+2.1900, +2.2077] | supported |
| heldout no condition minus conditioned profile tvd | +0.3138 [+0.3019, +0.3270] | supported |
| heldout no condition minus conditioned width absolute error | +0.9599 [+0.4548, +1.4967] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0352 [+0.0264, +0.0432] | supported |
| training conditioned minus independent reference conditional mmd | +0.0067 [+0.0066, +0.0068] | supported |
| training conditioned minus independent reference mean absolute error | +0.3687 [+0.3605, +0.3757] | supported |
| training conditioned minus independent reference profile tvd | +0.0933 [+0.0895, +0.0964] | supported |
| training conditioned minus independent reference width absolute error | +0.0278 [+0.0091, +0.0434] | supported |
| training no condition minus conditioned bitstring tvd | +0.2575 [+0.2402, +0.2765] | supported |
| training no condition minus conditioned conditional mmd | +0.0534 [+0.0513, +0.0554] | supported |
| training no condition minus conditioned mean absolute error | +2.2020 [+2.1924, +2.2116] | supported |
| training no condition minus conditioned profile tvd | +0.2440 [+0.2354, +0.2537] | supported |
| training no condition minus conditioned width absolute error | +1.1301 [+0.6214, +1.6714] | supported |

Table 17: Paired intervals for $n = 16$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | +0.0250 [+0.0190, +0.0313] | supported |
| heldout neighbor oracle minus conditioned conditional mmd | +0.0029 [+0.0026, +0.0032] | supported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.2461 [+0.2180, +0.2729] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0192 [-0.0205, -0.0178] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.1705 [-0.1893, -0.1523] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0559 [-0.0594, -0.0518] | unsupported |
| heldout mixture minus conditioned conditional mmd | +0.0001 [+0.0001, +0.0002] | supported |
| heldout mixture minus conditioned mean absolute error | -0.2429 [-0.2518, -0.2340] | unsupported |
| heldout mixture minus conditioned profile tvd | -0.0681 [-0.0696, -0.0666] | unsupported |
| heldout mixture minus conditioned width absolute error | -0.3812 [-0.4014, -0.3606] | unsupported |
| heldout no condition minus conditioned bitstring tvd | +0.2406 [+0.2034, +0.2781] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0405 [+0.0394, +0.0417] | supported |
| heldout no condition minus conditioned mean absolute error | +2.0498 [+2.0416, +2.0579] | supported |
| heldout no condition minus conditioned profile tvd | +0.1981 [+0.1780, +0.2176] | supported |
| heldout no condition minus conditioned width absolute error | +0.8736 [+0.4476, +1.2787] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0258 [+0.0219, +0.0293] | supported |
| training conditioned minus independent reference conditional mmd | +0.0064 [+0.0064, +0.0064] | supported |
| training conditioned minus independent reference mean absolute error | +0.4633 [+0.4530, +0.4731] | supported |
| training conditioned minus independent reference profile tvd | +0.0523 [+0.0495, +0.0553] | supported |
| training conditioned minus independent reference width absolute error | +0.1339 [+0.1182, +0.1497] | supported |
| training no condition minus conditioned bitstring tvd | +0.2100 [+0.1710, +0.2486] | supported |
| training no condition minus conditioned conditional mmd | +0.0499 [+0.0473, +0.0526] | supported |
| training no condition minus conditioned mean absolute error | +2.1053 [+2.0940, +2.1171] | supported |
| training no condition minus conditioned profile tvd | +0.1667 [+0.1570, +0.1770] | supported |
| training no condition minus conditioned width absolute error | +1.1659 [+0.7386, +1.5723] | supported |

Table 18: Paired intervals for $n = 19$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0610 [-0.0776, -0.0458] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | -0.0001 [-0.0003, +0.0001] | mixed |
| heldout neighbor oracle minus conditioned mean absolute error | +0.1758 [+0.1493, +0.2119] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0217 [-0.0240, -0.0192] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.1371 [-0.1597, -0.1162] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0175 [-0.0342, -0.0021] | unsupported |
| heldout mixture minus conditioned conditional mmd | +0.0003 [+0.0001, +0.0005] | supported |
| heldout mixture minus conditioned mean absolute error | +0.0533 [+0.0228, +0.0942] | supported |
| heldout mixture minus conditioned profile tvd | +0.0188 [+0.0164, +0.0213] | supported |
| heldout mixture minus conditioned width absolute error | +0.1641 [+0.1409, +0.1852] | supported |
| heldout no condition minus conditioned bitstring tvd | +0.2943 [+0.2769, +0.3101] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0389 [+0.0386, +0.0391] | supported |
| heldout no condition minus conditioned mean absolute error | +2.7073 [+2.6799, +2.7452] | supported |
| heldout no condition minus conditioned profile tvd | +0.3356 [+0.3282, +0.3468] | supported |
| heldout no condition minus conditioned width absolute error | +0.4266 [+0.1216, +0.9637] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0620 [+0.0488, +0.0763] | supported |
| training conditioned minus independent reference conditional mmd | +0.0065 [+0.0064, +0.0067] | supported |
| training conditioned minus independent reference mean absolute error | +0.3355 [+0.3104, +0.3554] | supported |
| training conditioned minus independent reference profile tvd | +0.0870 [+0.0847, +0.0893] | supported |
| training conditioned minus independent reference width absolute error | +0.0574 [+0.0395, +0.0769] | supported |
| training no condition minus conditioned bitstring tvd | +0.2394 [+0.2256, +0.2523] | supported |
| training no condition minus conditioned conditional mmd | +0.0533 [+0.0519, +0.0542] | supported |
| training no condition minus conditioned mean absolute error | +2.6746 [+2.6561, +2.6983] | supported |
| training no condition minus conditioned profile tvd | +0.2620 [+0.2594, +0.2645] | supported |
| training no condition minus conditioned width absolute error | +0.4747 [+0.1704, +1.0099] | supported |

Table 19: Paired intervals for $n = 19$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | -0.0606 [-0.0749, -0.0461] | unsupported |
| heldout neighbor oracle minus conditioned conditional mmd | +0.0002 [+0.0002, +0.0003] | supported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.1145 [+0.1000, +0.1286] | supported |
| heldout neighbor oracle minus conditioned profile tvd | +0.0016 [-0.0012, +0.0043] | mixed |
| heldout neighbor oracle minus conditioned width absolute error | -0.1451 [-0.1555, -0.1358] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0300 [-0.0443, -0.0155] | unsupported |
| heldout mixture minus conditioned conditional mmd | +0.0005 [+0.0005, +0.0006] | supported |
| heldout mixture minus conditioned mean absolute error | -0.0696 [-0.0804, -0.0585] | unsupported |
| heldout mixture minus conditioned profile tvd | -0.0029 [-0.0053, -0.0006] | unsupported |
| heldout mixture minus conditioned width absolute error | +0.0521 [+0.0416, +0.0622] | supported |
| heldout no condition minus conditioned bitstring tvd | +0.3037 [+0.2885, +0.3183] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0400 [+0.0394, +0.0404] | supported |
| heldout no condition minus conditioned mean absolute error | +2.6224 [+2.6098, +2.6350] | supported |
| heldout no condition minus conditioned profile tvd | +0.3360 [+0.3223, +0.3507] | supported |
| heldout no condition minus conditioned width absolute error | +0.8220 [+0.2164, +1.5105] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0428 [+0.0296, +0.0560] | supported |
| training conditioned minus independent reference conditional mmd | +0.0064 [+0.0063, +0.0064] | supported |
| training conditioned minus independent reference mean absolute error | +0.4279 [+0.4206, +0.4354] | supported |
| training conditioned minus independent reference profile tvd | +0.0878 [+0.0860, +0.0897] | supported |
| training conditioned minus independent reference width absolute error | +0.0319 [+0.0135, +0.0514] | supported |
| training no condition minus conditioned bitstring tvd | +0.2559 [+0.2370, +0.2723] | supported |
| training no condition minus conditioned conditional mmd | +0.0530 [+0.0507, +0.0549] | supported |
| training no condition minus conditioned mean absolute error | +2.6314 [+2.6155, +2.6474] | supported |
| training no condition minus conditioned profile tvd | +0.2668 [+0.2587, +0.2737] | supported |
| training no condition minus conditioned width absolute error | +1.0212 [+0.4158, +1.7108] | supported |

Table 20: Paired intervals for $n = 19$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

| Comparison | Effect [95% interval] | Status |
|---|---|---|
| heldout neighbor oracle minus conditioned bitstring tvd | +0.0074 [-0.0032, +0.0165] | mixed |
| heldout neighbor oracle minus conditioned conditional mmd | +0.0020 [+0.0016, +0.0023] | supported |
| heldout neighbor oracle minus conditioned mean absolute error | +0.2478 [+0.2051, +0.2986] | supported |
| heldout neighbor oracle minus conditioned profile tvd | -0.0111 [-0.0165, -0.0049] | unsupported |
| heldout neighbor oracle minus conditioned width absolute error | -0.1981 [-0.2308, -0.1643] | unsupported |
| heldout mixture minus conditioned bitstring tvd | -0.0656 [-0.0768, -0.0563] | unsupported |
| heldout mixture minus conditioned conditional mmd | +0.0002 [-0.0002, +0.0004] | mixed |
| heldout mixture minus conditioned mean absolute error | -0.2483 [-0.2763, -0.2077] | unsupported |
| heldout mixture minus conditioned profile tvd | -0.0619 [-0.0670, -0.0560] | unsupported |
| heldout mixture minus conditioned width absolute error | -0.4633 [-0.4961, -0.4286] | unsupported |
| heldout no condition minus conditioned bitstring tvd | +0.2272 [+0.1839, +0.2702] | supported |
| heldout no condition minus conditioned conditional mmd | +0.0401 [+0.0384, +0.0418] | supported |
| heldout no condition minus conditioned mean absolute error | +2.4747 [+2.4455, +2.5170] | supported |
| heldout no condition minus conditioned profile tvd | +0.2261 [+0.2088, +0.2418] | supported |
| heldout no condition minus conditioned width absolute error | +0.8519 [+0.1776, +1.5129] | supported |
| training conditioned minus independent reference bitstring tvd | +0.0385 [+0.0306, +0.0474] | supported |
| training conditioned minus independent reference conditional mmd | +0.0063 [+0.0062, +0.0066] | supported |
| training conditioned minus independent reference mean absolute error | +0.5383 [+0.5122, +0.5573] | supported |
| training conditioned minus independent reference profile tvd | +0.0603 [+0.0544, +0.0659] | supported |
| training conditioned minus independent reference width absolute error | +0.1626 [+0.1321, +0.1922] | supported |
| training no condition minus conditioned bitstring tvd | +0.1940 [+0.1476, +0.2405] | supported |
| training no condition minus conditioned conditional mmd | +0.0503 [+0.0465, +0.0542] | supported |
| training no condition minus conditioned mean absolute error | +2.5272 [+2.5060, +2.5506] | supported |
| training no condition minus conditioned profile tvd | +0.1807 [+0.1648, +0.1999] | supported |
| training no condition minus conditioned width absolute error | +1.1021 [+0.3924, +1.7942] | supported |

## H Archived Gaussian ablations

The topology, encoding, and ancilla studies predate the frozen factorial and are preserved as bounded ablations. Their metrics are regenerated from the archived JSON/NPZ pairs and are not pooled with the new campaign. Complete connectivity uses 190 phases at $n = 19$; the lattice and heavy-hex banks use 48 and 39. The symmetric-reupload comparison uses three matched seeds. The ancilla ladder uses $m \in \{0, 1, 2, 3, 5, 7\}$ and compares each augmented model with a visible-only matched-parameter control.

Permutation twirling replaces a raw sample $x$ by $\pi(x)$ for a uniformly random qubit permutation $\pi$. For an exchangeable target, this preserves the Hamming-weight profile and makes the output uniform within each observed shell in expectation. It is therefore a legitimate target-symmetry-aware post-processing baseline. It is not a symmetry learned by the circuit, it uses known target structure, and it is excluded from raw-circuit claims.

## I Negative two-coordinate feasibility study

An archived eight-qubit pilot attempted to condition jointly on Gaussian location and width. Its frozen design required single-condition representability before any shared-model or interpolation interpretation. At seed 0, the four single-condition bitstring TVDs were 0.3883, 0.3177, 0.2220, and 0.0945. The two narrow-width cells failed the declared capacity screen; their width errors were 0.4659 and 0.1058. The study stopped, so it supplies no evidence for two-coordinate conditional generation. We include it to mark the boundary of the present one-coordinate claim and to prevent selective reporting of only successful Gaussian variants.

## J Campaign integrity and reproduction

The Gaussian campaign is isolated under `results/gaussian-foundations-v1`. The immutable manifest digest binds the run plan; a source-tree digest binds the numerical core used by the resource preflight and every result, while a companion dependency-closure snapshot binds the encoder, ansatz, simulator, target, artifact, environment, and lockfile sources. Dataset archives record role, derived seed, condition grid, master-pool hash, and split hash. A result is complete only when its envelope, checkpoint, and probability archive all exist and their digests agree. Reservations without a complete optimizer state are never silently restarted.

From the repository root, the principal commands are:

```
uv sync --frozen
uv run python scripts/prepare_gaussian_campaign.py
uv run python scripts/run_gaussian_campaign.py preflight
uv run python scripts/freeze_gaussian_execution_source.py
uv run python scripts/run_gaussian_campaign.py run \
  --shard-count 5 --shard-index 0
uv run python scripts/summarize_gaussian_campaign.py
uv run python scripts/verify_gaussian_legacy.py
uv run python scripts/finalize_gaussian_claims.py
uv run python scripts/generate_gaussian_tables.py
uv run python scripts/plot_gaussian_campaign.py
uv run pytest -q
uv run ruff check src scripts tests
uv run python scripts/build_gaussian_arxiv_archive.py
uv run python scripts/audit_gaussian_release.py
uv run python scripts/run_curie_weiss_ancilla_campaign.py prepare
uv run python scripts/run_curie_weiss_ancilla_campaign.py preflight
uv run python scripts/run_curie_weiss_ancilla_campaign.py run \
  --shard-count 5 --shard-index 0
uv run python scripts/finalize_curie_weiss_ancilla_campaign.py
uv run python scripts/generate_curie_weiss_ancilla_tables.py
uv run python scripts/plot_curie_weiss_ancilla_campaign.py
uv run python scripts/audit_curie_weiss_ancilla_release.py
```

Shard indices 1–4 are executed analogously. Re-running a completed shard verifies and skips immutable outputs. Report, claim, table, and figure generators refuse partial campaign state.